MEA OCPP 1.6 Compliance Test Report

Section 4: Reset Check

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

2026-06-09 13:56

Contents

1 Test Configuration

Device Under Test	Vector vSECC.single Board
Device IP	192.168.1.166
OCPP Version	1.6 (JSON)
CP ID	rddQC4000001
CSMS	wss://ocpp.measandbox.com:2930
Connection	Direct (no proxy)
Test PC	192.168.111.185 (alias 192.168.1.200/24 on enp3s0)
EV Plug Timeout	0 s per plug event
Boot Wait Timeout	30 s (after Hard Reset)
Test Date	2026-06-09 06:36:03 UTC
Results file	tex/vsecc_section4_results.json

Test Architecture

Reset commands are issued via the MEA REST API (`POST /EV/remote/reset`). Three reset scenarios are tested:

1. **Hard Reset while idle (4.1–4.8):** Hard Reset is sent when the charger is idle. The vSECC reboots and sends a new BootNotification. A charging session is then started via RemoteStart.
2. **Soft Reset during charging (4.9–4.17):** Soft Reset is sent during an active charging session. The vSECC should gracefully send StopTransaction before rebooting. A new charging session is started after recovery.
3. **Hard Reset during charging (4.18–4.21):** Hard Reset is sent during an active charging session. The session may be aborted immediately; the vSECC reboots and reports Available after reconnection.

2 Section 4 Results

Summary: 6 PASS 0 FAIL 15 WARN 0 SKIP (21 total)

Item	Test Description	Detail / Evidence	Result
Flow 1: Hard Reset (Idle) — Items 4.1–4.8			
4.1	Reset (Hard) → Accepted	`HTTP 200`	PASS
4.2	BootNotification → Accepted (after Hard Reset)	`CSMS reconnection confirmed after reset`	PASS
4.3	StatusNotification Available (after reboot)	`Available not observed on MQTT in 20 s after reset`	WARN
4.4	StatusNotification Preparing (EV plug, Flow 1)	`EV_WAIT_SEC=0 | skipped` *Connect physical EV to connector and rerun*	WARN
4.5	RemoteStartTransaction → Accepted	`HTTP 200`	PASS
4.6	StartTransaction (triggered by RemoteStart)	`charging_session_state not observed on MQTT in 30 s`	WARN
4.7	StatusNotification Charging (Flow 1)	`Charging status not observed on MQTT in 20 s`	WARN
4.8	MeterValues (Flow 1, during charging)	`MeterValues not observed on MQTT in 15 s`	WARN
Flow 2: Soft Reset (During Charging) — Items 4.9–4.17			
4.9	Reset (Soft) → Accepted (during charging)	`HTTP 200`	PASS
4.10	StopTransaction (triggered by Soft Reset)	`session_state idle/finished not observed in 30 s` *Soft Reset should cause graceful StopTransaction before reboot*	WARN

Item	Test Description	Detail / Evidence	Result
4.11	StatusNotification Finishing (Soft Reset, EV plugged)	`Finishing not observed on MQTT in 15 s` *Some implementations skip Finishing → go directly to Available*	WARN
4.12	StatusNotification Available (unplug after Soft Reset)	`Available not observed on MQTT in 30 s | unplug EV`	WARN
4.13	StatusNotification Preparing (EV plug, Flow 2)	`EV_WAIT_SEC=0 | skipped` *Connect physical EV to connector and rerun*	WARN
4.14	RemoteStartTransaction → Accepted (Flow 2)	`HTTP 200`	PASS
4.15	StartTransaction (Flow 2, triggered by RemoteStart)	`charging_session_state not observed on MQTT in 30 s`	WARN
4.16	StatusNotification Charging (Flow 2)	`Charging status not observed on MQTT in 20 s`	WARN
4.17	MeterValues (Flow 2, during charging)	`MeterValues not observed on MQTT in 15 s`	WARN
Flow 3: Hard Reset (During Charging) — Items 4.18–4.21			
4.18	Reset (Hard) → Accepted (during charging)	`HTTP 200`	PASS
4.19	StopTransaction (triggered by Hard Reset)	`session_state idle/finished not observed in 30 s` *Hard Reset may not send StopTransaction before rebooting*	WARN
4.20	StatusNotification Finishing (Hard Reset, EV plugged)	`Finishing not observed on MQTT in 15 s` *Hard Reset typically skips Finishing — goes directly Available after reboot*	WARN
4.21	StatusNotification Available (after Hard Reset, EV unplug)	`Available not observed in 30s | unplug EV / wait for reboot`	WARN

3 Notes on Failures and Warnings

Reset command

The MEA sandbox REST API endpoint `POST /EV/remote/reset` accepts `"reset_type": "Hard"` or `"reset_type": "Soft"`. Items 4.1, 4.9, and 4.18 are PASS if the API returns HTTP 200.

Hard Reset reboot timing

After a Hard Reset the vSECC physically reboots. The test waits up to `BOOT_WAIT_SEC` seconds for the MQTT `vsecc/ocpp_connection_status=connected` event or a successful REST API probe to confirm reconnection.

EV-dependent items

- **4.4, 4.13** StatusNotification Preparing: require a physical EV to plug in within the `EV_WAIT_SEC` timeout.
- **4.6–4.8, 4.15–4.17** Charging session items depend on RemoteStart succeeding with an EV present.

StopTransaction on reset

- **4.10** Soft Reset: the OCPP specification requires the charger to send StopTransaction with `reason=SoftReset` before restarting.
- **4.19** Hard Reset: the specification allows the charger to immediately reboot without a graceful Stop-Transaction; session is recovered on the CSMS side via a timeout or explicit cleanup.

4 Dependency Map

Flow 1 4.1 (Reset Hard) → 4.2 (Boot) → 4.3 → 4.4 → 4.5 → 4.6 → 4.7 → 4.8

Flow 2 4.9 (Reset Soft) → 4.10 → 4.11 → 4.12 → 4.13 → 4.14 → 4.15 → 4.16 → 4.17

Flow 3 4.18 (Reset Hard) → 4.19 → 4.20 → 4.21